10.2 DCGANs

- previous attempts to scale up GANs using CNNs to model images have been unsuccessful
- however, after extensive model exploration, it has been identified a family of architectures that resulted in stable training across a range of datasets and allowed for training higher-resolution and deeper generative models
- core to the approach is adopting and modifying three previous demonstrated changes to CNN architectures

10.2 DCGANs

- the first is the all-convolutional network, which replaces deterministic spatial pooling functions (such as max-pooling) with strided convolutions, allowing the network to learn its own spatial downsampling
- this approach will be used in the generator, allowing it to learn its own spatial upsampling, and also in the discriminator

10.2 DCGANs

- second is the trend towards eliminating fully-connected layers on top of convolutional features
- the strongest example of this is global average pooling, which has been utilized in image classification models
- global average pooling was found to increase model stability, but hurt convergence speed
- a middle ground of directly connecting the highest convolutional features to the input and output, respectively, of the generator and discriminator worked well
- the first layer of the GAN, which takes a uniform noise distribution $\boldsymbol{z}$ as input, could be called fully-connected, as it is just a matrix multiplication, but the result is reshaped into a 4-dimensional tensor and used as the start of the convolutional stack
- for the discriminator, the last convolutional layer is flattened and then fed into a single sigmoid output
- see Figure 2 for a visualization of an example model architecture

10.2 DCGANs

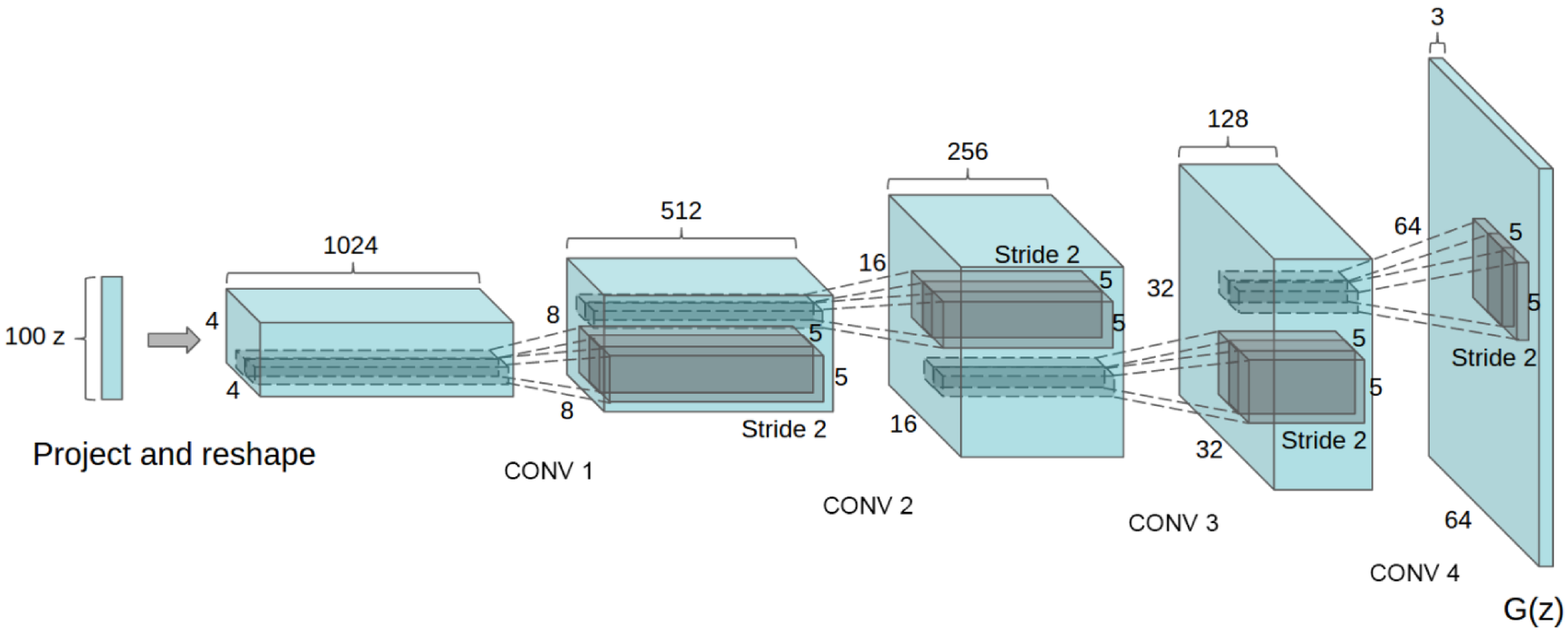

Figure 2: DCGAN generator used for LSUN scene modeling. A 100-dimensional uniform distribution $\boldsymbol{z}$ is projected to a small spatial extent convolutional representation with many feature maps. A series of four transposed convolutions then convert this high-level representation into a 64×64-pixel image. Notably, no fully-connected or pooling layers are used.

10.2 DCGANs

- third is batch normalization, which stabilizes learning by normalizing the input to each unit to have zero mean and unit variance
- this helps deal with training problems that arise due to poor initialization and helps gradient flow in deeper models
- this proved critical to get deep generators to begin learning, preventing the generator from collapsing all samples to a single point, which is a common failure mode observed in GANs
- directly applying batch normalization to all layers, however, resulted in sample oscillation and model instability
- this was avoided by not applying batch normalization to the generator output layer and the discriminator input layer

10.2 DCGANs

- the ReLU activation is used in the generator, with the exception of the output layer, which uses the tanh function
- it has been observed that using a bounded activation allowed the model to learn more quickly to saturate and cover the color space of the training distribution
- within the discriminator, it was found the that Leaky ReLU works well, especially for higher-resolution modeling
- this is in contrast to the original GAN paper, which used the maxout activation

10.2 DCGANs

- to conclude, following are architecture guidelines for stable Deep Convolutional GANs:
  - Replace any pooling layers with strided convolutions (discriminator) and transposed convolutions (generator).
  - Use batch normalization in both the generator and the discriminator.
  - Remove fully-connected hidden layers for deeper architectures.
  - Use ReLU activation in generator for all layers except for the output, which uses tanh.
  - Use LeakyReLU activation in the discriminator for all layers.
- samples from the DCGAN model trained on the LSUN bedrooms dataset containing a little over 3 million training examples can be seen in Figure 3

10.2 DCGANs

Figure 3: Generated bedrooms after five epochs of training on the LSUN bedrooms dataset. There appears to be evidence of visual underfitting via repeated noise textures across multiple samples, such as the base boards of some of the beds.